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The Impact of Bank Consolidation on Credit Supply and Performance

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Between 2009 and 2011, the Spanish banking system underwent a restructuring process based on savings banks' consolidation. The program's design allows us to study how banks' consolidation affects credit supply and performance. We propose a quasi-experimental analysis showing that bank mergers restrict credit supply and set higher interest rates but also reject fewer applicants and report fewer nonperforming loans. We then estimate a structural model of credit in which banks set interest rates and lending standards. We find that, despite the relaxation in their lending standards, merged banks' credit performance improved thanks to a significant drop in their screening costs. (*JEL* G21, G34, L22)

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Fierce competition can give rise to excessive risk taking in banking systems featuring many undiversified banks. If bad risks translate into problematic loans, public intervention drawing on government funds may become necessary to rescue troubled banks. A structural policy that regulators often consider to solve the problems of over-banked systems consists of fostering bank consolidation. This happened in Europe, where, in the past decade, the banking sector of many countries was significantly affected by restructuring measures (European Commission 2017). It also happened in the United States, where the FDIC auction process was used after the crisis to resolve insolvent banks, equivalent to a regulator-induced consolidation process (Granja et al. 2017; Allen et al. 2024).

The literature has demonstrated that postmerger, banks tend to curtail their credit supply, particularly affecting small and medium-sized enterprises (see, among others, Berger et al. 1998; Peek and Rosengren 1998; Sapienza 2002; Bonaccorsi di Patti and Gobbi 2007; Degryse et al. 2011). This literature has also explored the potential offset of these costs through the efficiency gains produced by consolidation (Houston et al. 2001; Focarelli and Panetta 2003; Panetta et al. 2009; Erel 2011), or mergers' capacity to mitigate the propagation of financial shocks (Petersen and Rajan 1995; Scharfstein and Sunderam 2016; Favara and Giannetti 2017; Giannetti and Saidi 2019). We contribute to this debate by quantifying the effects of bank consolidation on both the supply and the performance of credit. We combine quasi-experimental methods based on a novel identification strategy and an equilibrium model of bank competition. We propose the first structural model in the literature that allows consolidated banks to adjust both interest rates and lending standards in equilibrium. This allows us to determine the impact of consolidation on credit volumes, lending standards, interest rates, expected defaults, and profitability.

We first put forward an instrumental variable identification strategy to show that, on the one hand, merged banks restrict credit supply and increase interest rates applied to new loans; on the other hand, they reduce the rejections of new borrowers' loan applications and report fewer nonperforming loans (NPLs). We then propose a new equilibrium model of borrowers' demand for credit from differentiated banks. The model shows that bank consolidation produces a drop in the quantity of credit, resulting from the increase in interest rates applied to new loans, despite the rise in the acceptance rates and expected default risk. The fall in credit quantity for merged banks suggests that the negative impact of the increase in interest rates on the amount of credit is not offset by the relaxation of credit standards. We also find that merged banks report a substantial fall in the costs of screening borrowers, which capture, in a reduced-form way, the costs associated with accurately assessing borrower credit merits or the process of firm monitoring. Overall, the reduced quantity of credit, the increase in interest rates, and the drop in screening costs imply that merged banks' profits increase despite extending loans to riskier borrowers.

Based on these results, we conclude that credit performance improves with consolidation.

To conduct our empirical analyses, we use data from the credit registry of the Banco de España for the years between 2007 and 2011. Our policy experiment is the Spanish savings banks consolidation program (the program from now on). In the years before the 2008 crisis, head-to-head competition and weak governance led savings banks to make poor investment choices, as witnessed by hoarding credit to the construction sector, ultimately leading to an NPL problem. The government gave troubled savings banks the possibility of obtaining public capital in exchange for the submission of a consolidation plan. Between November 2009 and December 2010, the program led to a consolidation wave in which the number of savings banks went from 37 to 12. In practice, all savings banks participated in consolidation operations. The asset value of the institutions participating in the program amounted to about 1,300 billion Euro (BE), a figure comparable to the combined total value of U.S. M&A transactions across all industries during 2009 and 2010.[1]

The Spanish consolidation program is an ideal laboratory to conduct our empirical analysis. The Banco de España Central Credit Registry gives us access to comprehensive data that can be used to identify the effects of consolidation on credit volumes, lending standards, interest rates, and credit performance. We have access to information on the outstanding amount of firms' credit with each bank at the monthly level and the volume of monthly NPLs reported by banks on each firm. We also have information on firms' probability of default and banks' requests for information regarding the credit situation of new borrowers seeking a loan. These data are commonly used in the literature as a valid proxy for loan applications (e.g., Jiménez et al. 2014), which allows us to identify banks' acceptance or rejection decisions. Finally, the information on the interest rate banks apply to newly issued loans is more aggregate and available only at the bank-month level.

Another advantage of the Spanish consolidation program is that it allows us to perform a quasi-experimental analysis, which we use to motivate the structural model. Savings banks' consolidation occurred through either standard mergers or the establishment of business groups. Merged banks had a centralized organizational structure, while business group banks retained their status as stand-alone legal entities, reporting credit operations independently to the regulator. Our descriptive analysis shows that business group banks were observationally comparable to merged banks. Still, their decentralized structure hindered their ability to coordinate lending practices and exercise market power to the extent possible in a full-fledged merger. Thus, by comparing merged and business group banks, we can identify the impact of mergers' market power on credit supply and performance.

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1 See www.statista.com/statistics/420990/value-of-merger-and-acquisition-deals-usa/.

This analysis is conducted using an instrumental variable approach. Our instrument for the decision on the form of consolidation is based on factors that are arguably exogenous to savings banks' financial and economic characteristics. We find such an instrument in the political pressure exerted by regional governments on savings banks to consolidate via a merger with other savings banks in the same region, as opposed to a business group. We use savings banks' market-share overlap in 1995 to proxy for the political pressure exerted by regional governments on local savings banks. We use 1995 because, at the time, savings banks were mainly located around their primary area of influence, or headquarters. Compared to business groups, we find that merged banks reduced credit supply, increased interest rates on newly issued loans, and reduced the rejections applied to new borrowers' loan applications. Additionally, they reported a significantly lower fraction of NPLs.

A limitation to interpreting these results is that the quasi-experimental analysis cannot identify the equilibrium forces driving the restriction in credit supply. On the one hand, we find that the credit performance extended by merged banks improved. On the other hand, the contemporaneous increase in interest rate spreads and the reduction in rejections applied to new borrowers could reflect that merged banks were extending credit to more risky borrowers at the extensive margin. Unfortunately, the quasi-experimental analysis alone cannot determine whether the reduction in NPLs comes from a reduction in the credit extended by merged banks to risky borrowers or from merged banks becoming more efficient in screening risky firms or the banks improving their monitoring practices to reduce delinquencies. To progress on this front, we need an equilibrium model of bank competition.

In our model, differentiated banks engage in a Bertrand-Nash competition. Banks' credit supply is jointly determined by the interest rates and the acceptance rates that they optimally set in equilibrium (with and without consolidation). Since the model allows us to unpack the equilibrium forces producing the changes in the outcomes of interest postconsolidation, in the structural analysis, we focus on the effects of bank mergers. We estimate the demand and supply sides of the model using data spanning the periods both before and after savings banks' consolidation. On the demand side, the comprehensive data from the Spanish credit registry allows us to compute accurate estimates of borrower probability of default and price elasticity, utilizing information on banks' requests for information on new borrowers to estimate the probability that their application is accepted. On the supply side, we quantify how banks change their interest and acceptance rates with consolidation. We use the model's equilibrium conditions to recover heterogeneous costs across banks, both before and after consolidation, which allows us to determine whether cost efficiencies arise during our sample period. Utilizing the model's estimates and equilibrium assumptions, we simulate counterfactual scenarios where we divest merged banks while varying the extent of cost efficiencies. Thanks to this model, we can analyze the effects of mergers on credit volumes, interest rates, lending standards, expected defaults, and profitability.

In line with the quasi-experimental analysis, the model predicts that bank mergers restrict the quantity of credit. Merged banks set higher interest rates and acceptance rates compared to what they do when they are not consolidated. The market power effect pushes the interest rates set by merged banks up. Higher interest rates raise merged banks' profit margins, enabling them to tolerate an increase in their acceptance rates despite lending to borrowers with more significant default risk. The net effect of this is a decrease in the quantity of credit for merged banks because the impact of the substantial increase in interest rates on credit supply dominates the higher acceptance rates.

These results raise the question of whether the increase in the risk taken by merged banks through the increase in acceptance rates causes an increase in merged banks' profitability. The model allows us to show that mergers significantly drop banks' screening costs, equivalent to reducing the cost of extending credit to borrowers of any risk level. We find that merged banks' lower screening costs, with the drop in the supply of credit and increased interest rates, offset the increase in total costs due to the relaxation in lending standards. Consequently, profits increase by approximately 22% for merged banks. We conclude that bank mergers enhance the performance of the credit extended in the economy. Despite the different approach, these results are consistent with the reduced-form findings that merged banks increase interest rates, reduce quantity of credit and NPLs and relax lending standards. Bolstering our interpretation of the structural results, we find that the volume of NPLs reported by merged banks is lower than that of business group banks regardless of firms' probability of default. Notably, the decrease in NPLs is larger for ex ante riskier firms, which indicates that, compared to business group banks, more effective screening by merged banks post consolidation allowed them to take on more risk without compromising the credit quality of their portfolios.

We contribute to the growing literature on the structural modeling of financial markets, developing an equilibrium model of credit demand and supply based on the workhorse demand model for differentiated products from empirical industrial organization (see Clark et al. 2021 for a review). Several papers in this field estimate structural models of credit demand and loan pricing using contract-level data (Crawford et al. 2018; Benetton, 2021; Robles-Garcia, 2022). We instead rely on market share level data (similarly to, e.g., Aguirregabiria et al. 2024; Wang et al. 2022), because we observe interest rates only at the bank-month level, and complement information on the market shares with data on loan applications and rejections. Thanks to the information available on loan applications, we model banks' decisions on interest and acceptance rates in the spirit of Crawford et al. (2019). To our knowledge, ours is the first model that analyses how banks' loan pricing and lending standards affect credit supply and how banks' decisions change with consolidation.

Only a few other papers have used loan application data to estimate a structural model of credit demand and supply. Cuesta and Sepulveda (2021) use this information to model borrowers' choice to apply for a loan, whereas we use the rejection information to model banks' decision to reject borrowers. Lim (2020) models banks' interest rates and acceptance rates in the U.S. mortgage market but does not consider how consolidation affects these margins. Thanks to our model we show that, with consolidation, restrictions in credit supply by merged banks are accompanied by a relaxation in lending standards through the increase in the acceptance rates applied to more risky borrowers. Our formulation of banks' cost structure allows us to also quantify the cost savings due to consolidation by distinguishing between those coming from reductions in operating costs and those coming from reductions in screening costs. By combining the analysis of credit outcomes and cost efficiencies, our counterfactual analysis documents an improvement in credit performance through a reduction in the value of banks' costs.

Modeling the choice on banks' acceptance rates introduces choice-set heterogeneity across borrowers based on their degree of risk, in the spirit of Sovinsky Goeree (2008) and Abrams (2019). Different from those papers, though, we can leverage the wealth of data in the Spanish credit registry to estimate the parameters governing the acceptance probability. The combined data on banks' requests for information on loan applications submitted by new borrowers and credit exposure to firms allows us to trace the decision taken by the bank on the application, as we can observe whether a new credit relationship is established with the borrower following the request for information.

We also contribute to the empirical literature that studies the impact of bank mergers on credit, cited at the beginning of this introduction. We propose a quasi-experimental analysis that exploits the institutional features of the Spanish consolidation program to identify the impact of bank mergers on credit supply and performance. We offer a comprehensive analysis of how merger consolidation changes credit supply, through adjustments to the price of credit and the lending standards set by the banks. We also quantify the impact of changes in credit supply on loan performance by looking at the change in NPL accumulation. The value added of the structural analysis consists of unveiling the equilibrium forces driving the results obtained in the quasi-experimental analysis. It allows us to quantify how changes in interest rates and lending standards affect equilibrium demand and supply of credit at the borrower level.

Finally, we contribute to the literature studying the effects of imperfect competition in selection markets, both theoretically (Lester et al. 2019) and empirically in insurance (Starc, 2014) and credit markets (Adams et al. 2009; Einav et al. 2012; Allen et al. 2014). Relative to this body of empirical work, we are the first to provide evidence of the effect of a countrywide bank consolidation program on the composition of borrowers. We show that, on the one hand, bank mergers take more risk by relaxing their lending standards; on the other hand, they become more efficient in screening borrowers. Overall, the combination of these effects results in an improved performance of the credit they extend.

1. The Savings Banks' Sector Consolidation Program

In the aftermath of the financial crisis that hit the country in 2008, the Spanish government enacted the Royal Decree 9/2009 (*Real Decreto-Ley 9/2009*) of June 26, 2009 (the Law from now on), to restructure the sector of Spanish savings banks (*cajas de ahorros*). At the time, savings banks' assets represented about 40% of Spanish banking assets (European Commission, 2017). Tough competition in a highly fragmented market and weak governance practices often led savings banks to make poor investment choices. For example, as of 2010, Spanish savings banks were exposed to the construction sector for 217BE, of which about 100BE were problematic.

The Spanish government did two things to restructure the sector. First, it allowed savings banks to obtain public capital from a special fund (*Fondo de Reestructuración Ordenada Bancaria*, or FROB) in exchange for the submission of a consolidation plan. Savings banks were asked to submit a plan giving details on the operation of consolidation and how they intended to restructure their business. The savings banks that did not present severe financial difficulties could simply submit a consolidation plan to strengthen their capital position without applying for FROB's intervention. Second, the Law reformed the corporate governance of all savings banks, including by placing them into private ownership. The goal was to reduce the political influence exerted by the ownership of regional governments up to November 2009.

The Law allowed savings banks to consolidate via an M&A or a *sistema institucionales de protección* (SIP). SIPs are a form of business group. Their structure had to conform to the prescriptions of the Law and was thus uniform across groups and provinces within groups. SIPs feature similarities with standard M&As, and one crucial difference. We start with the former. First, in analogy to M&As, SIP banks were required to establish pacts of full mutual assistance on liquidity and solvency and were responsible on a consolidated basis for fulfilling regulatory requirements. Second, the Law requires that SIPs last at least 10 years. Third, like M&A banks, SIP banks had access to consolidated information on the firms interacting with other savings banks in the same group, so SIP banks did not need to tap Banco de España for this information. Finally, M&A and SIP banks were subject to the same capital regulation and NPL reporting obligations as all other Spanish banks.

The key difference between M&A and SIP banks is that the latter remained separate legal entities. This means they kept their consolidation brands and continued reporting their credit operations independently to the credit registry.

This makes the organizational structure of SIPs less centralized than that resulting from M&As. SIP banks' legal independence is likely to impair the coordination of credit policies within the group due to the possibly different use of the credit-merit analyses produced by the risk management central unit depending on loan officers' incentives. Finally, being legally separate, SIP banks were still subject to antitrust enforcement and thus viable for infringing antitrust laws if they were to collude.

The consolidation program went fast, bringing the number of savings banks from 37 to 12 between November 2009 and December 2010.[2] The program featured full compliance: savings banks accounting for about 90% of the credit extended in the sector participated in an operation of consolidation. Since the program targeted the savings bank sector, no such bank was consolidated with a commercial or a cooperative bank during the period under consideration.

The decision between M&As and SIPs was influenced by considerations related to regional politics. During the early phase of the program, all M&A activities occurred among savings banks with headquarters located within the same region. Regional governments opposed cross-region M&As out of concern that the headquarters of the newly merged bank might relocate outside the region (Banco de España, 2017). In response to these political concerns, the Constitutional Court clarified that the primary objective of the program was to promote the stability of the financial system (Méndez Álvarez-Cedrón, 2011). Consequently, the Banco de España encouraged savings banks to form SIPs instead (Banco de España, 2017). SIPs allowed savings banks to consolidate while maintaining their legal independence.

Consolidating savings banks into business groups, akin to the Spanish SIP model, is common in Europe. In countries such as Germany, Norway, and Sweden, there are business group structures where savings banks keep their brands while entrusting certain functions to a central unit. Although the specific delegated functions may vary across groups, a shared objective is to achieve synergies by pooling essential resources, such as risk management, and by realizing cost savings.[3]

2. Data and Descriptive Statistics

We primarily use data from the Banco de España Central Credit Registry (CCR). The CCR contains detailed monthly information on the credit position of each Spanish firm with each Spanish bank for all loans above 6,000

[2] Table A-1 in the Internet Appendix reports the list of the operations of consolidation (SIPs and M&As) that took place during this period, which are those we consider in our empirical analysis.

[3] The German *Sparkassen-Finanzgruppe* is a financial group composed of legally independent savings banks that cooperate to achieve cost savings. In Norway, most savings banks formed alliances, like the *Sparebank 1 Alliance* and the *Eika Group*, within which, among other things, they share the processing of information on borrower credit merit. Finally, the Swedish *Swedbank* is an alliance of savings banks that covers information technology and payment transactions.

euros, including credit lines. Thus, we observe the virtual universe of bank exposures to nonfinancial corporations. We aggregate the information on the credit amount extended by each bank to each firm every month to get the total credit, including both drawn and undrawn amounts for credit lines. This feature allows us to trace all the changes in credit flows between a given bank and a given firm over time. The CCR contains data on the reported volume of NPLs related to specific firms. However, although we can compute the ratio of NPLs over total loans at the bank-firm level, we do not have enough information to identify the specific loans that are nonperforming. We also observe the requests for information made by banks on firms' credit situation. In line with previous research (e.g., Jiménez et al. 2014), we assume that these firms seek a bank loan from the banks that submit a request for information about them. Given that the CCR contains information on the outstanding credit balances, by restricting the sample to those firm-bank pairs with no credit exposures when the applications are submitted, we can infer whether or not the firm obtained the loan from the bank that requested information.[4] Finally, the interest rates applied to new loans are obtained from the Banco de España supervisory data. During our sample period, loan interest-rate information is available only at the bank-month level.

Since the CCR reports the identifier of each bank and firm, we merge the bank-firm level data with the balance sheets of banks and firms. The Banco de España collects the data on banks in its role of banking supervisor. The data are used to obtain proxies for bank size (logarithm of total assets), leverage (total liabilities over total assets), and profitability (return on assets, or ROA).

The information in the CCR is also merged with the dataset of the Spanish nonfinancial firms that respond to the Integrated Central Balance Sheet Data Office Survey (CBSDO), which contains information from the accounts filed with the mercantile registries for more than 850,000 firms in all years between 2006 and 2010 (as of the version of the data set available in March 2020). This data set includes information on firms' identifiers, the industry of operation, and other items of the balance sheet that enable us to obtain proxies for firms' size, leverage, and profitability (constructed analogously to those for the banks) and liquidity (liquid assets over total assets).

Finally, we use the information on the FROB funds transferred to each bank, obtained from the FROB webpage, to assist with the consolidation.

3. The Effect of Mergers' Market Power on Credit Supply and Performance

To establish the impact of M&A consolidation on credit supply and performance, we need a group of banks whose market power does not change

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[4] Our analysis includes observations of successful and unsuccessful applications submitted to banks requesting CCR information. However, our data do not allow us to identify some potential unsuccessful applications, namely, those to banks that do not request information from CCR and those to banks with a previous relationship with the firm, which we removed from our sample. Note that banks receive monthly information from CCR about their current customers.

after consolidation but who are otherwise similar to M&A banks. We take advantage of the institutional design of the Spanish consolidation program and compare M&A and SIP banks. This section provides descriptive evidence on credit conditions, performance, and operating costs for M&A and SIP banks. We then show that SIP and M&A groups differ in the extent to which they adopt centralized lending policies. Finally, we describe the empirical strategy we follow in our quasi-experimental analysis and present our findings.

3.1 Sample construction and descriptive statistics

The sample period goes from November 2007 to November 2011. The data set comprises information on 527,739 firm-bank relationships and 383,546 nonfinancial corporations (368,812 preconsolidation and 270,870 postconsolidation). We consider the 37 savings banks participating in one of the 12 M&A or SIP operations between November 2009 and December 2010. These savings banks account for about 40% of the total credit in the economy and 90% of the total credit in the savings banks' sector. The list of operations is in Table A-1 in the Internet Appendix. We trace the effects of these consolidations up to November 2011. We stopped in November 2011 because, during the next quarters, Spain received rescue packages to cope with the European sovereign debt crisis.

The individual savings banks that formed a M&A group ceased their operations between November 2009 and December 2010 to function as a single entity. SIP banks continued to report individual information to the credit registry. Consequently, after consolidation, we use the group j-level information available for M&As and aggregate the information for the savings banks belonging to each SIP. Before consolidation, we aggregate the information for the savings banks that will later be part of M&As or SIPs. To this aim, we aggregate each balance sheet item (total assets, total liabilities, and total income) of all credit institutions that are part of each new banking group and then obtain the corresponding ratios.

Panel A of Table 1 reports information on savings banks' characteristics over the sample period. As in our regression analysis, the information is aggregated at the bank group level. Panel B reports information on firm characteristics during the sample period. It shows that the firms in our data are small, with an average asset value of about 2ME (corresponding to the asset-based threshold for small firms according to the European Commission Recommendation 2003/361/EC).

In panels C and D, we report information on the semiannual growth rate of total credit volume, the proportion of NPLs over total loans at the end of each semester, the monthly interest rate spread, and the rejected applications in a given semester before and after consolidation. Credit growth and NPL proportion data are aggregated at the bank-firm-semester level. The information on interest rate spreads is at the bank-month level. They are computed as the difference between nominal interest rates and the 3-month

Table 1
Summary statistics: Quasi-experimental analysis

A. Bank groups

Variables	November 2007–November 2011					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
TA (BE)	83.13	47.73	83.40	19.73	280.46	60
Capital ratio (%)	5.60	4.94	1.71	3.67	8.88	60
ROA (%)	0.42	0.45	0.28	0.05	0.79	60

B. Firms

Variables	November 2007–November 2011					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
TA (ME)	1.90	0.47	5.42	0.04	6.95	1,035,318
Total liabilities/TA (%)	72.74	79.90	26.29	19.14	100.00	1,035,318
Liquid assets/TA (%)	9.34	3.32	14.86	0.00	41.59	1,035,318
ROA (%)	3.86	5.13	19.05	−24.54	27.74	1,035,318

C. Bank-firm relationships

Variables	November 2007–Consolidation date					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
Δ *log(Credit)*	−0.05	−0.02	1.38	−2.40	2.77	1,666,166
NPL (%)	1.80	0.00	13.03	0.00	0.00	1,666,166
Rejected applications (Count)	0.42	0	0.59	0	1	174,696
Interest rate spread (%)	1.84	1.61	0.88	0.67	3.39	354

D. Bank-firm relationships

Variables	Consolidation date–November 2011					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
Δ *log(Credit)*	−0.13	−0.04	1.19	−2.26	1.06	936,838
NPL (%)	3.09	0.00	17.02	0.00	0.00	936,838
Rejected applications (Count)	0.27	0	0.51	0	1	68,449
Interest rate spread (%)	2.73	2.78	0.58	1.71	3.56	234

This table contains descriptive statistics (mean, median, standard deviation, 5th and 95th percentiles, and number of observations [N]) for bank and firm characteristics (panels A and B, respectively) as well as for firm-bank credit balances (panels C and D). Panel A reports end-of-year information at the level of each individual group j, panel B at the level of individual firms-year, and panels C and D are at the bank-firm-semester level for credit change, number of rejected applications, and NPLs, and at the bank-month level for interest rate spreads. Panels A and B report the statistics between November 2007 and November 2011. Panel C reports descriptive statistics on the semiannual change in the credit balance between November 2007 and the date in which consolidation took place, on the number of rejected applications received by a firm from the savings banks within a given group in each semester, on the ratio of NPLs over total loans for the whole sample of firm-bank pairs at the end of each semester, and the average monthly interest rate spread over the 3-month Euribor at the bank-month level. Panel D does the same for the period between the date of consolidation and November 2011. Firm-level variables and the log change in credit are winsorized such that we set the observations above (below) the 99% (1%) percentiles at the value of the 99% (1%) percentile. For additional information on the construction of these variables, see the Internet Appendix.

Euribor (Euro Interbank Offered Rate). To determine the count of rejected loan applications, we assume that a firm credit application to a given bank is rejected if we do not observe an increase in the firm-bank outstanding credit balance in the month the firm applies for credit or within the following 3 months. We aggregate the number of rejections of all savings banks within a given group to a given firm on a semiannual basis. We exclude from the sample the observations related to the semesters during which a given bank participates in an operation of consolidation. This ensures coherence in the bank group's structure when applications occur and decisions are made. Additionally, we drop the outliers in the number of information requests submitted by a bank in a given semester.[5]

The panels show that the total volume of credit decreased more in the postconsolidation period than in the preconsolidation period. The ratio of NPLs over total assets and the interest rate spreads increased between the pre- and postconsolidation periods. Finally, the number of rejected applications exhibits a downward trend.

3.2 Descriptive evidence on lending practices

We next present descriptive evidence comparing M&A and SIP banks. We will show that compared to SIP banks, M&A banks reduce their credit supply and report improved credit performance. We also show that M&A and SIP banks feature similar levels of operating costs before and after consolidation. Finally, we provide evidence consistent with weaker coordination of credit policies within SIP groups, as opposed to M&A groups.

3.2.1 Credit supply and operating costs. In Figure 1, we plot the pattern of the following variables: the average amount of outstanding credit granted to the universe of nonfinancial corporations (top-left panel), the average spread between nominal interest rates and the 3-month Euribor (top-right panel), the value of the ratio between the volume of NPLs and banks' total loans (bottom-left panel), and the ratio between yearly operating costs and the value of operating costs in 2007. We do this separately for our sample of M&A and SIP banks.

The evolution of the new credit granted by M&A and SIP banks (top-left panel) follows a comparable pattern before November 2009. In the 2 years before November 2009, M&A banks extended between 15BE and 20BE more credit than SIP banks. Starting from the second semester after November 2009, the sign of the difference reverts. By the end of the fourth semester after November 2009, M&A banks extended approximately 10BE less credit than SIP banks. That the change in the differential effect starts one semester after the Law was passed is to be expected, as most M&A operations occurred at the end of the first quarter and during the second quarter of 2010.

The pattern of interest rate spreads in the top-right panel mirrors that of total credit. Before November 2009, M&A and SIP banks' spreads followed a

[5] There are months in which a few banks request information on an unusually high number of firms. Keeping these observations in the sample could lead to a problem of mismeasurement. We classify these specific bank-month pairs as outliers when the number of loan applications in a given month is (1) more than 10 times larger than the average number of monthly applications of that bank over our sample period and (2) higher than the average number of applications of that bank plus two times the standard deviation in the bank number of applications.

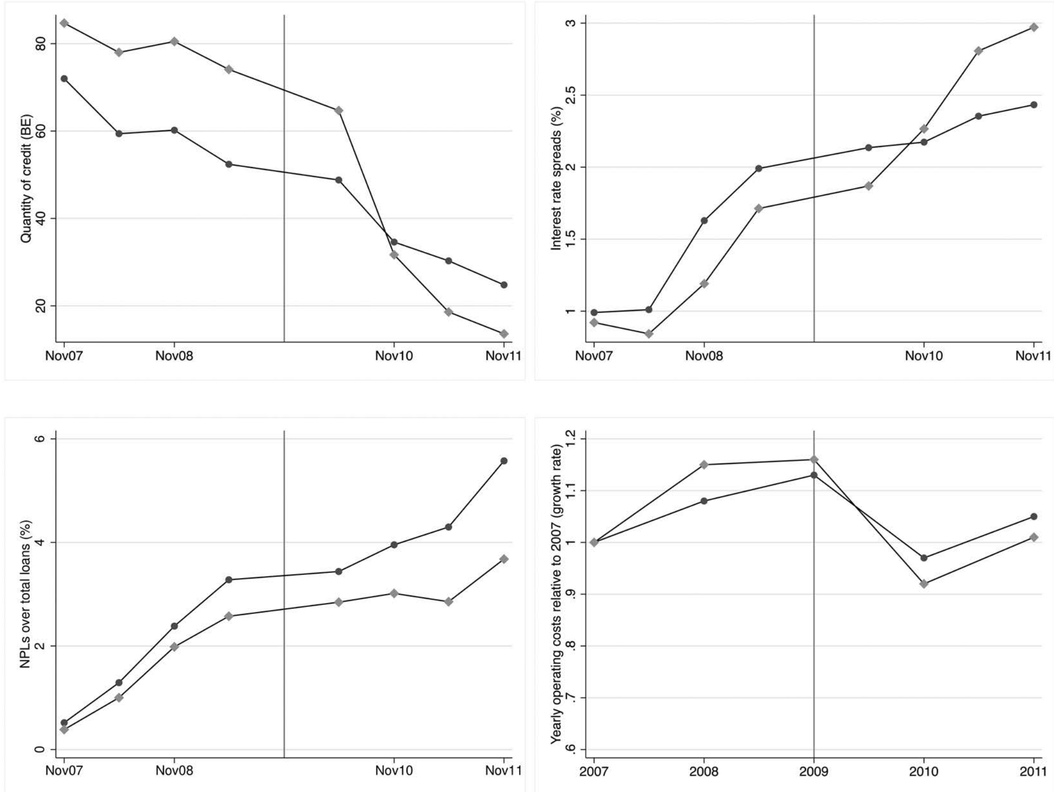

Figure 1
Unconditional evidence
In the top-left, top-right, and bottom-left panels, we report the semiannual pattern of the quantity of credit, the interest rate spreads relative to the 3-month Euribor, and the NPLs over total loans for M&A banks (grey line, squares) and SIP banks (black line, circles) in the period ranging between four semesters before and four semesters after the month in which the Law was passed (November 2009). In the bottom-right panel, we report the yearly operating costs relative to 2007 for M&A banks (grey line, squares) and SIP banks (black line, circles) in the 2 years before and the 2 years after 2009. Since the information is aggregated at the group level, we do not plot the confidence intervals. For information on the construction of the variables, see the Internet Appendix.

common trend. On average, M&A banks applied lower spreads than SIP banks before November 2009 and until the second semester after November 2009. Then, contemporaneously with the reversion in the patterns of total credit, SIP banks applied cheaper average spreads.

In the bottom-left panel, we observe a common trend in the pattern of the NPL ratio for M&A and SIP banks during the 2 years before the program. Starting in the second semester after November 2009, the ratio slowed significantly more for M&A than for SIP banks. Finally, possibly because of the start of the European sovereign debt crisis, the NPL ratio of M&A and SIP banks spiked in the fourth semester after the beginning of the program.

The bottom-right panel shows that M&A and SIP banks featured similar operating costs before and after consolidation. When comparing the value of yearly operating costs relative to 2007 at the group level, M&A and SIP banks follow a comparable pattern. The postconsolidation data show a relatively small reduction in operating costs for both bank groups. This evidence has two takeaways. First, the formation of SIP banks results in operating cost reductions similar to those of M&A banks, at least in the short run.[6] This aligns with the Spanish regulator's expectation that SIPs generate efficiencies comparable to M&A banks.[7] Second, the evidence confirms previous findings indicating that operating cost reductions postconsolidation take time to occur (Focarelli and Panetta, 2003). This is likely because banks need to undertake lengthy operational adjustments, like branch closures, to achieve such savings.

On top of the banks' operating costs documented in this section, the structural analysis in Section 4 allows us to estimate the screening costs that banks bear when setting their lending standards. Arguably, the latter can be adjusted more quickly than operating costs by changing banks' credit policies related to screening or monitoring borrowers. Instead, operating costs require structural adjustments. Confirming this intuition, the structural analysis will document reductions in savings banks' screening costs postconsolidation for M&A banks.

3.2.2 Coordination of credit policies. We now provide evidence suggesting that SIP banks engage in a weaker centralization and coordination of credit policies than M&A banks. We utilize data from the credit register of Banco de España, as this information offers insights into banks' processing of borrower information and their responses to new loan applications. This analysis allows us to assess whether there are differences between M&A and SIP banks in the extent to which they coordinate their decisions.

6 The results are similar when comparing the number of bank branches or the yearly ratio between operating costs and income.

7 In the words of the Banco de España former deputy governor (Javier Aríztegui): "SIPs are expected to produce the same organizational improvements, efficiencies, economies of scope, diversification, and quality as traditional M&As. They must do this within the same period as a classic merger and must put all the necessary efforts into making these results be perceived by the market as permanent" (December 2010).

According to the theory proposed by Stein (2002), loan officers in more centralized organizations tend to rely more on hard information when determining lending conditions than do loan officers in decentralized organizations. As M&A groups are inherently more centralized than SIP groups, we expect that M&A banks show a greater reliance on hard information compared to SIP banks. To verify this hypothesis, we examine the data on banks' requests for credit history information of new loan applicants. These data align with the definition of hard information in Liberti and Petersen (2019), as they are verifiable, easy to store and transmit, and their contents are independent of their collection. By computing the number of information requests submitted by each group (M&A and SIP) from the consolidation period until November 2011 and normalizing it by the total credit granted by each group, we find that M&A banks have a 15% higher propensity, per thousand euros of credit granted, to access the credit registry and assess the past performance of loan applicants.

Like M&A banks, postconsolidation SIP banks were compelled by the Law to set up a new central risk management system. However, the decentralized structure of SIP groups may imply a different use of the credit-merit analyses produced by the risk management unit across the banks in the group, depending on loan officers' incentives. This can impair the coordination of lending policies within the group. To test whether this intuition is correct, we again use the data on requests for information on new borrowers.

We construct a dummy variable that is equal to 1 if (1) a savings bank belonging to a given SIP requested information on a borrower in any month after the group's formation and (2) we do not observe an increase in the credit balance with that borrower during this period. The dependent variable is then equal to 0 when a rise in the firm-bank credit balance follows the request for information, which we interpret as a successful loan application. We regress this variable on dummy variables for each specific savings bank and on SIP group-firm fixed effects. Using these fixed effects implies that our sample is restricted to those observations for which two savings banks within a given SIP group request information on the same new borrower during the period under consideration. This allows us to effectively capture whether a given firm is treated likewise by the savings banks within the same SIP. If all savings banks within a SIP treat a firm's loan application similarly, the individual savings bank dummy variables should not be statistically significant.

Table 2 shows that a savings bank treats a loan application submitted by the same firm differently than the average savings bank in the same SIP in four out of the six SIP groups. Given that each SIP has a different number of savings banks, we report the coefficient only for the savings bank dummy with the lowest *p*-value within each SIP group to guarantee confidentiality.

These pieces of descriptive evidence motivate our assumption in the structural model that only M&A banks take coordinated credit policies.

Table 2
Uncoordinated lending policies across SIP banks

	Rejected application
Savings bank in SIP group 1	0.333
	[0.320]
Savings bank in SIP group 2	0.432***
	[0.114]
Savings bank in SIP group 3	0.467***
	[0.111]
Savings bank in SIP group 4	0.324***
	[0.062]
Savings bank in SIP group 5	0.338
	[0.270]
Savings bank in SIP group 6	0.141**
	[0.069]
Observations	1,518
R-squared	.677
SIP Group-Firm FE	YES

In this table, we test whether savings banks within a given SIP have similar lending policies after forming the group. We restrict our sample to the savings banks in our sample that formed an SIP. The analysis was conducted between the month of the formation of SIP group j and November 2011. Our dependent variable is a dummy variable equal to 1 if (1) a savings bank belonging to a given SIP requested information on a firm at any month after the group's formation and (2) we do not observe an increase in the firm-bank credit balance during this period. When these conditions are jointly satisfied, we infer that the firm loan application was rejected. We regress this variable on dummy variables for each specific savings bank (the omitted term refers to one of the savings banks in each given SIP) and on SIP group-firm fixed effects. We report the coefficient only for the savings bank dummy with the lowest p-value. We report robust standard errors in brackets. $^{*}p<.1$; $^{**}p<.05$; $^{***}p<.01$.

3.3 Identification strategy

Our strategy to identify the impact of M&A banks' market power on credit supply and performance relies on the comparison between M&A and SIP groups. As shown above, SIP banks are comparable to M&A banks in the operating costs they bear before and after consolidation but are less capable of implementing coordinated lending policies.[8] Since coordination is a key condition for the exercise of market power, the differences in credit supply and performance across M&A and SIP banks should reflect M&A groups' stronger market power.

However, if the decision to form a M&A as opposed to a SIP is explained by unobserved factors that are correlated with savings banks' inclination to restrict credit, by, for example, implementing a stricter screening of borrowers, then the comparison between M&As and SIPs would not necessarily capture the effects of interest. We need an instrument for the decision on the form of consolidation based on factors that are arguably exogenous to savings banks' financial and economic characteristics. We find such an instrument in the political pressure exerted by regional governments on savings banks to consolidate via M&As with other savings banks in the same region.

[8] In Table A-2 (Internet Appendix), we show that, before consolidation, M&A and SIP banks were also comparable with respect to a battery of balance sheet characteristics, including profitability, capitalization, and NPL ratios.

In 1989, Spain lifted the regulation banning the opening of savings banks' branches across local areas. As of 1995, about 80% of the savings banks were still in their original market (Fuentelsaz and Gomez, 2001). Thus, location in 1995 highly correlates with savings banks' proximity to their primary area of influence or headquarters.[9] We then use savings banks' market-share overlap in 1995 to proxy the political pressure exerted by regional governments on local savings banks. Accordingly, we rely on information about savings banks' credit overlap to construct the instrument for the decision to do an M&A between November 2009 and December 2010. The logic is as follows: if political pressure to do a within-region M&A explains the consolidation mode, then the savings banks that do an M&A during the consolidation program are those whose branches are more likely to overlap in their area of influence in 1995.

We use a measure of overlap based on the credit extended by the second-largest savings bank in group j over the total credit extended by all the savings banks in province m,

$$Credit\ Overlap_{jm} = \frac{Total\ Credit_{b'm}}{\sum_{b \in m} Total\ Credit_{bm}}, \tag{1}$$

where b' is the second largest savings bank in group j and market m. Although this measure may underestimate the extent of overlap when considering groups composed of equally large savings banks, it is a particularly suitable proxy in applications like ours, where M&As and SIPs are heterogeneous in the number and size of the savings banks in each group.

With mergers between two banks, a natural measure of overlap would consider the market share of the smallest bank (Erel, 2011). In our setting with consolidations between more than two banks, taking the market share of the smallest bank may result in a downward bias in overlap. For instance, when there are three savings banks, the third entity may have a significantly smaller market share than the others, yet substantial overlap could exist between the two largest banks in the province. Alternatively, with mergers between equally sized banks, one could use the total market share of the resulting group as a proxy for overlap. However, this measure would not differentiate between consolidations among banks with asymmetric market shares. Consider the following two mergers: in the first, a bank with a 49% market share merges with one with a 1% market share, while in the second, two banks with 25% market shares merge. If we were to rely on a measure based on the total market share, both groups would have the same level of overlap. Using the second-largest market share allows us to distinguish between these two situations.

In column 1 of Table 3, we regress the binary value $M\&A_j$, which is equal to 1 if group j forms through an M&A, on credit overlap, defined as in Equation (1); bank controls; and province fixed effects. We use credit data

[9] January 1995 is the first month in which credit information is available in the Spanish credit registry.

Table 3
Credit overlap and consolidation form

Variables	(1) All regions	(2) All regions	(3) "Main" regions
Credit overlap in 1995	3.823***	3.535***	4.289***
	[1.278]	[0.983]	[1.305]
Observations	330	330	55
R-squared	.464	.432	.364
Province FE	YES	NO	NO
Bank Controls	YES	YES	YES

The table reports the results of the regressions that relate the dummy variable that is equal to 1 for M&As and 0 for SIPs and a measure of credit overlap in 1995 at the province level as defined in Equation (1). In columns 1 and 2, we use all the regions with provinces in which the savings banks were active in 1995, whereas in column 3 we use province-level data from the "main" region of each group. In all the columns, we use the following set of bank controls: *log(TA)*, *Capital ratio*, *ROA*, and *(FROB funds)/TA*. Moreover, in column 1, we use province fixed effects. Robust standard errors (in brackets) are clustered at the province level. $^{*}p<.1$; $^{**}p<.05$; $^{***}p<.01$.

from the provinces (m) in which the savings banks were active in 1995.[10] In column 2, we drop province fixed effects. The positive and significant coefficients confirm that savings banks with a larger market overlap in their area of influence in 1995 were disproportionally more likely to consolidate via an M&A between November 2009 and December 2010. Since, as of 1995, savings banks were mainly located in their primary area of influence, this result is consistent with our reconstruction of how the consolidation program unfolded in Section 1, with regional governments trying to veto across-region M&A to limit the risk that the headquarters of the new bank moves out of the region.

To instrument the choice of the consolidation form, we need to aggregate the overlap measure at the group level. We want the value of our instrument to capture the higher likelihood that credit overlap in the region of influence, the "main" region, is larger for M&A banks. We proceed as follows. All M&A operations occur between banks having their headquarters in the same region, so we use the information on the credit overlap of M&A groups using data from that region. In the case of SIPs, we consider the region where the group has the largest credit exposure. After selecting the relevant "main" region for M&A and SIP groups, we compute overlap defined as in Equation (1) for the provinces within that region. We do this because, for antitrust purposes, the relevant market in banking is defined at the province level (Guiso et al. 2004). In column 3, we confirm the positive relationship between the M&A dummy and credit overlap in 1995 using province-level data from each group's "main" region j.[11]

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10 Spain has 17 autonomous regions and two autonomous cities: Ceuta and Melilla. Each autonomous region is divided into provinces, making a total of 50 plus Ceuta and Melilla.

11 In column 3, we do not include province fixed effects. Their use would reduce the sample to 30 observations, given that there are some provinces for which we have information on a single group once we restrict the sample

Finally, for our instrument, we take the maximum value of province-level overlap within each group's "main" region. This gives us one credit overlap value per bank group. Subsequently, we create a binary variable, taking the value of 1 if the maximum value of credit overlap exceeds the median and 0 otherwise. We use the resulting binary variable to instrument the decision on the consolidation form in our regressions. The correlation between the instrument and the instrumented variable (the binary variable M&A) is around 0.7. Moreover, our instrument takes a value of 1 (indicating a high-credit overlap) in the "main" regions of five of the six savings banks' groups that merged through M&As.

3.4 Estimation and results

We now present our empirical results on the differential effects of M&A consolidation on the supply and performance of credit,[12] based on instrumental variable regression analyses in which we instrument M&A consolidation through a binary variable that indicates whether the credit overlap in the bank group region of influence is high (as explained in Section 3.3).

3.4.1 Credit volumes, lending standards, and interest rate spreads. Consider a bank group j dealing with firm i at time t (corresponding to a semester). The baseline econometric model we use for the analysis of bank credit is:

$$\begin{aligned} y_{jit} = {} & \alpha_1 M_j + \alpha_2 Post_{jt} + \alpha_3 (M_j \times Post_{jt}) \\ & + \beta X_{jt-h} + \gamma Z_{it-h} + \zeta FROB_{jt} + \delta_{kmst} + \epsilon_{jit}. \end{aligned} \tag{2}$$

We denote by y_{jit} the semiannual growth rate of the (log) volume of total credit between November 2007 and November 2011. $Post_{jt}$ is specific to each bank group j and is equal to 1 starting from the end of the semester in which group j (M&A or SIP) was formed. $M\&A_j$ is a binary variable that equals 1 if the savings bank's group j formed through a M&A. The coefficient of interest is α_3. It captures how consolidation differentially affected the outcome variable for M&A banks relative to SIP banks. All the specifications are estimated including predetermined control variables (X_{jt-h} and Z_{it-h}). X_{jt-h} contains a bank's *Total assets (TA)*, *Capital ratio*, and *ROA*. Z_{it-h} contains firm *Leverage*, *Liquidity*, *ROA*, and *TA*. The values of the variables in X_{jt-h} and Z_{it-h} are taken at the end of the year before the beginning of each semester in which we compute the semiannual growth rate. Finally, $FROB_{jt}$ denotes the value of FROB's capital injections received by bank group j between 2009 and 2011.

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to the "main" region per group. However, as shown in columns 1 and 2, using province-fixed effects does not change our results.

12 In the main text, we report our results using the instrumental variable methodology described in Section 3.3. We obtain analogous results when using ordinary least squares (OLS; see Table A-3 in the Internet Appendix).

To control for firm-specific shocks, we use industry (k), location (m), size (s), and time (t) fixed effects (δ_{kmst}). This means we exploit the variation arising from the credit conditions applied to firms of the same size, in terms of assets' decile, and within the same period, NACE industry, and province.[13]

The specification in Equation (2) is estimated by an instrumental variable regression with high-dimensional fixed effects. We cluster standard errors at the industry-province-size and bank group j-province level. We consider a two-way clustering that includes the interaction between bank group j and province because, to study bank competition, the relevant local market is typically considered the province.

Column 1 of Table 4 reports the estimates of Equation (2). Compared to SIP banks, we find that M&A banks extend less credit.[14] The estimate in column 1 implies that, compared to SIP banks, M&A banks cut lending by about 3% on a semiannual basis, or about 16,000 euros per firm per semester.[15] We perform the standard Kleibergen-Paap Rank LM statistic to check whether the instrumental variable specification is identified, that is, whether the excluded instrument is correlated with the endogenous regressor. We reject the null hypothesis that the equation is underidentified; thus, the instrument is relevant, and the model is identified.

The result in column 1 is consistent with M&A banks restricting credit supply because of the market power effect they gain postconsolidation. This outcome can arise from banks raising their interest rates along the supply curve or a contemporaneous adjustment of their lending standards. We study this next.

We first consider the differential effect of M&A consolidation on lending standards. We propose a regression analysis in which the dependent variable is the number of rejected applications by the savings banks within a given group to a given firm in a semester. The instrumental variable and the explanatory variables are otherwise the same as in Equation (2). We restrict the sample to those firm-bank pairs with no credit exposures when the application is submitted. The results are reported in column 2 of Table 4. We observe that postconsolidation, M&A banks reject 0.111 fewer applications than SIP banks, corresponding to 25% of the average number of applications at the firm-semester level rejected over our sample period. This result suggests

[13] Degryse et al. (2019) show that industry-location-size-time fixed effects are more appropriate for controlling demand differences relative to firm-time fixed effects. Using the latter, we would restrict the sample of firms to consider only those that take credit from multiple banks during the sample period.

[14] Despite the use of industry-location-size-time fixed effects, the coefficient on *Post* can be estimated in columns 1–3 because the associated dummy is defined at bank-time level, and in the first semester not all savings banks were consolidated.

[15] In the Internet Appendix, Figure A-1 shows the dynamic effects of the consolidation program. The left panel plots the coefficients obtained running our specification for the growth rate of total credit volume. We find no trend in the point estimates before November 2009. After November 2009, the coefficients become negative and significant. The right panel plots the coefficients obtained running our specification for the NPLs in the semesters postconsolidation. All the coefficients are negative and significant.

Table 4
The effect of market power on credit supply and performance

	(1) Δ *log(Credit)*	(2) *# Rejected applications*	(3) *Interest rates*	(4) *NPLs*
Post	0.001	–0.031	0.265	
	[0.025]	[0.046]	[0.234]	
M&A	–0.010	–0.055*	–0.447	–0.011***
	[0.021]	[0.033]	[0.361]	[0.004]
Post × *M&A*	–0.032*	–0.111**	0.758*	
	[0.019]	[0.056]	[0.351]	
Observations	2,603,004	243,145	588	121,003
Industry-Location-Size-Time FE	YES	YES	–	YES
Bank Controls	YES	YES	YES	YES
Firm Controls	YES	YES	–	YES

This table reports the results obtained from three instrumental variable regression analyses in which the instrument is a binary variable that indicates whether the credit overlap in the bank group region of influence is high. The dependent variable in column 1 is the semiannual growth rate of the (log) volume of total credit between November 2007 and November 2011. Column 2 shows the results obtained from a regression analysis analogous to that in column 1 but using as the dependent variable the number of rejected applications received by a firm from the savings banks within a given group j in each semester. Column 3 reports the results obtained from a regression analysis in which the dependent variable is the spread of the average monthly interest rate charged by a given bank group j to new loans granted in month t to nonfinancial institutions over the 3-month Euribor. The information on interest rates is available for different categories of loan maturity (less than 1 year, between 1 and 5 years, and more than 5 years). We perform a regression analysis in which the interest rate is the weighted average across the three maturity buckets, using the new operations within each maturity bucket as weights so that the unit of observation is bank-month. In columns 1 to 3, the sample period goes from November 2007 to November 2011. In column 4, we report the results obtained in a regression analysis in which the dependent variable is the average proportion of NPLs at the end of every semester in the period spanning between the announcement of group j-formation (M&A or SIP) and November 2011. The explanatory variable of interest in columns 1 to 3 is the interaction of a dummy variable that is equal 1 if the savings bank chooses an M&A (and 0 if they choose an SIP) and a dummy variable that equals 1 starting from the end of the semester in which group j (M&A or SIP) formed. Since column 4 considers the bank-firm pairs with no credit relationship at the date of consolidation, the variable of interest is the dummy M&A, which is estimated only for the postconsolidation period. The set of control variables in all columns includes bank characteristics such as *log(TA)*, *Capital ratio*, *ROA*, and *(FROB funds)/TA*. Additionally, in columns 1, 2 and 4 we use industry-location-size-time fixed-effects and the following firm characteristics as control variables: *Total liabilities/TA*, *Liquidity/TA*, *ROA*, and *log(TA)*. Robust standard errors (in brackets) are clustered at industry-province-size and bank group-province level in columns 1, 2 and 4 and at the bank group and month level in column 3. $^{*}p<.1$; $^{**}p<.05$; $^{***}p<.01$. For information on the construction of the variables, see the Internet Appendix.

that M&A banks relax their lending standards applied to new borrowers postconsolidation. The Kleibergen-Paap Rank LM statistic confirms that the excluded instrument is not correlated with the endogenous regressor.

We now analyze the differential impact of M&A consolidation on interest rate spreads applied to new loans. We use the following empirical specification:

$$w_{jt}=\alpha_1 M_j+\alpha_2 Post_{jt}+\alpha_3(M_j\times Post_{jt})+\beta X_{jt-h}+\zeta FROB_{jt}+\iota_{jt}, \quad (3)$$

where w_{jt} denotes the spread between the nominal interest rate and the 3-month Euribor. The dependent variable is constructed by taking the weighted average of the interest rate across three maturity buckets using as weights the new credit operations within each bucket ($<$1 year, 1-5 years, and $>$5 years). Since, during our sample period, the information on interest rates applied to new loans by each bank is collected by the regulator at the monthly level, in this regression, the unit of observation is at the bank-month level. $Post_{jt}$ is equal to 0 between

November 2007 and the month in which group-j consolidation took place and 1 from the month of consolidation to November 2011. The specification includes the same bank controls (X_{jt-h}) used in Equation (2), FROB contributions ($FROB_{jt}$), and no firm controls. Clustering is at the bank-group j and month level, corresponding to the level of aggregation of the information on interest rates.

The results of our instrumental variable regression are in column 3 of Table 4. Compared to SIP banks, M&A banks apply higher interest rate spreads post consolidation.[16] Back-of-the-envelope calculations based on the coefficient in column 3 suggest that a loan granted by an M&A bank is 31 bp more expensive than that given by an SIP bank postconsolidation. This means that the premium charged by M&A banks corresponds to 9% of the average baseline spread with the 3-month Euribor rate (3.3%). These results align with the reduction in the quantity of credit documented in column 1. Also, based on the Kleibergen-Paap Rank LM statistic, we reject the null hypothesis that the equation is underidentified.

The reduction in credit documented in column 1 is consistent with the increase in the interest rate spreads in column 3. The contemporaneous decrease in the number of rejections applied to new borrowers suggests that M&A banks are also changing the composition of borrowers at the extensive margin. The quasi-experimental analysis cannot determine whether the new borrowers are riskier than those who receive credit in the baseline. To perform this type of analysis, we need to model how the change in interest rates affects individual borrowers' demand for credit on the demand side. We also need to model banks' cost structure on the supply side to determine whether, postconsolidation, a relaxation in lending standards is associated with a worsening in the quality of borrowers or whether consolidation produces efficiencies in the cost of screening borrowers. We perform this analysis using the equilibrium model of bank competition given in Section 4.

3.4.2 NPL accumulation. For the analysis of NPLs, we use the following specification:

$$z_{jit} = \alpha M_j + \beta X_{jt-h} + \gamma Z_{it-h} + \zeta FROB_{jt} + \delta_{kmst} + \epsilon_{jit}. \tag{4}$$

The dependent variable is the proportion of NPLs over total loans of a given firm i reported by a bank group j at the end of each semester between the date of group-j consolidation and November 2011. We consider only the firm-savings bank pairs that, within a group j, feature no credit

[16] We find similar results when using the interest rate corresponding to each maturity bucket, so that the unit of observation is at the bank-month-maturity level, and estimate the coefficients of interest using a weighted OLS regression. We also run the specification for interest rate spreads by using the share of new loans in each of the three maturity buckets in our data set as a dependent variable. The consolidation form (M&A or SIP) has no differential impact on new loans' maturity.

relationship at the beginning of the consolidation process (i.e., November 2009). Consequently, there is no $Post_{jt}$ dummy. We do this because we cannot identify the specific loan facility that is nonperforming in the post period. If we were to consider the firms with a relationship with a savings bank before consolidation, some NPLs reported postconsolidation may be associated with lending originating before consolidation. The main advantage of this approach is that loan refinancing, or evergreening, cannot impair the interpretation of our findings. The specification contains the same firm and bank controls (Z_{it-h} and X_{jt-h}, respectively) used in Equation (2) and controls for the value of FROB contributions ($FROB_{jt}$). We cluster standard errors at the industry-province-size and bank group j-province level.

The results of the instrumental variable regression are in column 4 of Table 4. We find that M&A banks report a smaller proportion of NPLs than SIP banks. The estimates imply that the share of firm credit that turns out to be nonperforming is about 1.1 pp less for M&A banks than for SIP banks, representing a decrease of approximatively 35% in the average NPL ratio observed in panel D of Table 1 (3.09%). Given that some loans are unlikely to have matured within the postconsolidation period we consider, ours is a lower bound to the long-term effect of consolidation on NPLs. Finally, the Kleibergen-Paap Rank LM statistic results confirm that the instrument is relevant and the model is identified.

To follow up on the discussion of the results in columns 1–3, the quasi-experimental evidence on NPL accumulation could come from M&A banks reducing credit to risky borrowers, extending more credit to risky borrowers, and contemporaneously becoming more efficient in screening these firms or improving monitoring to minimize delinquencies. In the next section, we use the model to quantify the savings in screening costs generated by M&A consolidation.

4. Equilibrium Model of Banks' Competition

We propose a structural analysis to quantify the implications of the consolidation program for credit supply and performance. We estimate a new equilibrium model of borrowers' demand for credit from differentiated banks and borrowers' default. On the supply side, banks engage in a Bertrand-Nash competition. Banks' credit supply is jointly determined by the interest rates and the acceptance rates that they optimally set in equilibrium (with and without consolidation). This allows us to assess the importance of modeling banks' decisions on their lending standards when evaluating the effects of consolidation in credit markets.

We use the model's estimates and equilibrium assumptions to simulate counterfactual scenarios in which we divest consolidated banks and vary the extent of cost efficiencies. In the quasi-experimental analysis, we use the group of SIP banks to control for the operating efficiencies generated by consolidation. Our structural model can unpack the equilibrium forces behind the changes in credit supply and performance produced by M&A banks. We first quantify the impact of the market power effect of M&A consolidation on equilibrium outcomes like interest rates, lending standards, borrowers' default risk, and quantity of credit, absent any merger-related efficiency. Then, we quantify the total impact of these forces on the conditions and the credit performance extended by M&A banks. Last, we document the importance of jointly modeling banks' interest rates and lending standards.

In the model, we allow bank market shares to vary at the bank-province-month level, which is the natural unit of analysis of bank competition. Since the credit registry contains information on interest rates at the bank-month level, we can analyze interest rates and lending standards using only the information on prices and quantity aggregated at the bank-month level. This approach aligns with much of the structural literature in industrial organization. Using aggregate bank-month level interest rates, as opposed to borrower-level ones when the latter data is not available is also a common practice among papers estimating structural models of demand for loans (Aguirregabiria et al. 2024; Wang et al. 2022). The lack of information on interest rates at the borrower level also means that we cannot model risk-based pricing. We compensate for this by allowing banks to screen borrowers based on expected borrowers' default risk aggregated at the bank-month level.

4.1 Model

We take as the unit of observation a bank $b=1,...,B_{mt}$ in a province $m=1,...,M$ at a month $t=1,...,T$. We assume that borrower i's demand for loans is determined by the following indirect utility function:

$$U_{ibmt}^{D}=\underbrace{\alpha^{D}P_{bt}+\beta^{D}X_{bt}+\tau_{b}^{D}+\omega_{mt}^{D}+\xi_{bmt}^{D}}_{\equiv\delta_{bmt}^{D}}+\varepsilon_{ibmt}^{D}. \tag{5}$$

P_{bt} is the average interest rate paid by borrower i on bank's b new loans in month t, X_{bt} is a time-varying measure of bank size (logarithm of total assets, *log(TA)*), τ_b^D are bank fixed effects, ω_{mt}^D are province-month fixed effects, ξ_{bmt}^D are unobserved (by the econometrician) bank-province-month attributes, and ε_{ibmt}^D are IID Type-1 Extreme Value shocks. We allow borrowers to select an outside option, whose indirect utility is normalized to 0, that we define as a set of small fringe banks.[17]

Banks are differentiated firms that compete Bertrand-Nash to attract borrowers on interest rates P_{bt} and a parameter γ_{bt}, whose value captures the

[17] Fringe banks represent small local banks, with an average combined share across markets of around 11%. We do not assume that the outside option corresponds to the value of not borrowing for two reasons. First, we do not have records in the data of firms that do not take credit; we observe only borrowing firms. Second, even though we observe borrowing firms that stop borrowing, if we wanted to assume that they chose an outside option of not borrowing, we would need to make an assumption on what their potential borrowing capacity could have been, which is unobserved in the data.

stance of banks' lending standards. Screening in our context implies that each bank b at time t decides whether borrower i will be accepted based on the following probability:

$$\phi_{ibt} = \frac{\exp(\gamma_{bt} + \lambda_{ibt})}{1 + \exp(\gamma_{bt} + \lambda_{ibt})}, \tag{6}$$

where λ_{ibt} is a borrower-bank-month component and γ_{bt} is the bank-month specific parameter that determines the acceptance rates that banks change to tighten or loosen their lending standards. The model implies that by setting a higher value of the lending standards' parameter γ, the bank accepts more applicants.

Differently from Sovinsky Goeree (2008), who models heterogeneous choice sets in a way that is similar to what we do, we can estimate the parameters of the acceptance probability ϕ directly, as we observe acceptances and rejections from the datasets on banks' requests for information regarding loan applications submitted by new borrowers. However, given that we observe applications only for the subsample of new borrowers, we aggregate acceptance probabilities at the bank-month level ϕ_{bt} and define banks' market shares as:

$$s_{bmt} = \sum_{\mathcal{S} \in \mathcal{C}_{bt}} \prod_{l \in \mathcal{S}} \phi_{lt} \prod_{k \notin \mathcal{S}} (1 - \phi_{kt}) \frac{\exp(\delta^D_{bmt})}{1 + \sum_{r \in \mathcal{S}} \exp(\delta^D_{rmt})}, \tag{7}$$

where $\mathcal{C}_{bt}$ is the set of all choice sets that include bank b at month t, and $\mathcal{S}$ represents each individual choice set. The demand probabilities depend on both the interest rate P, through δ^D_{bmt}, and the lending standards' parameter γ, through ϕ_{bt}. Intuitively, a higher value of the parameter γ_{bt} increases the value of ϕ_{bt} which, in turn, raises both borrower demand and the value of a bank's market share (s_{bmt}).

Banks form expectations about borrowers' default on their loans, based on the following borrower indirect utility from defaulting:

$$U^F_{ibt} = \alpha^F P_{bt} + \beta^F X_{bt} + \tau^F_b + \omega^F_t + \xi^F_{bt} + \varepsilon^F_{ibt}, \tag{8}$$

where the determinants of the default utility are analogous to those in Equation (5), and ε^F_{ibt} are IID Type-1 Extreme Value shocks. A positive α^F captures the idea that, conditional on banks' lending standards determined by Equation (6), an increase in interest rates can attract a riskier pool of borrowers. This means that a bank's share of defaulting loans in month t is:

$$F_{bt} = \frac{\exp(\alpha^F P_{bt} + \beta^F X_{bt} + \tau^F_b + \omega^F_t + \xi^F_{bt})}{1 + \exp(\alpha^F P_{bt} + \beta^F X_{bt} + \tau^F_b + \omega^F_t + \xi^F_{bt})}. \tag{9}$$

In the baseline, banks' equilibrium interest rate (P_{bt}) and lending standards' parameter (γ_{bt}) are determined by maximizing expected profits:

$$\Pi_{bt} = [(1 + P_{bt})(1 - F_{bt}) + R_{bt} F_{bt} - C_{bt}] Q_{bt}, \tag{10}$$

where $Q_{bt} = \sum_m^M s_{bmt} \mathcal{M}_{mt}$ corresponds to the loan amount granted by bank b at time t, $\mathcal{M}_{mt}$ is the total potential amount that could be borrowed in a province-month combination, R_{bt} is the recovery rate on defaulted loans,[18] and C_{bt} are average costs, specified as follows:

$$C_{bt} = \exp(c_{0bt} + c_{1bt}\gamma_{bt}). \tag{11}$$

They indirectly depend on the part of the quantity of credit through the lending standards parameter γ_{bt}, which affects s_{bmt} (see Equations (6) and (7)).

Our model allows for two channels of credit risk affecting banks' expected profits. Ex ante, banks must formulate expectations regarding firms' default risk implied by adjusting their interest rates for a given level of lending standards. Accordingly, we expect higher interest rates to increase the value of firms' ex ante risk of default F. Ex post, while we interpret c_{0bt} as a proxy for banks' baseline operating costs (e.g., those related to managing branches or workers), the value of c_{1bt} corresponds to how variations in banks' lending standards through the parameter γ_{bt} influence their average costs. In practice, c_{1bt} captures, in a reduced-form way, the expenses linked to screening borrowers. This process might involve accurately assessing borrower creditworthiness or monitoring firms over time. If c_{1bt} is positive, setting a higher parameter γ (i.e., accepting more loan applications) leads to lending to riskier firms, and the additional cost incurred for extending credit to riskier borrowers is represented by the positive value of c_{1bt}. In what follows, we refer to c_{1bt} as the screening cost component of average costs.

Our key contribution to the banking literature consists of modeling two of the margins through which banks typically adjust their supply of credit and how they affect banks' cost structure. Interest rate P operates at the intensive margin (i.e., along the supply curve). By changing interest rate P, the bank increases or decreases the quantity of credit for given lending standards γ, and internalizes the effect on expected borrower default F. The parameter γ operates at the extensive margin. Through γ, the bank can change the quality of its borrowers in a way that depends on the sign of c_{1bt}. If the estimate of c_{1bt} is has a positive sign, an increase in the value of γ rotates the cost curve inward. Consequently, the bank optimally lends to riskier borrowers, increasing the lending cost. If, as it happens in our setting, consolidation gives rise to a contemporaneous reduction in screening costs c_{1bt}, the effect of an increase in γ on costs is ambiguous.

Similarly to Crawford et al. (2019), we can back out the banks' average cost parameters (c_{0bt} and c_{1bt}) by using the first-order conditions with respect to the

[18] Based on banks' balance sheet information, we set the recovery rate at 55%.

interest rate and the screening parameter. The first-order conditions imply that:

$$\frac{\partial \Pi_{bt}}{\partial P_{bt}} = [(1+P_{bt})(1-F_{bt})+R_{bt}F_{bt}-C_{bt}]\frac{\partial Q_{bt}}{\partial P_{bt}} + [1-F_{bt}+(R_{bt}-1-P_{bt})\frac{\partial F_{bt}}{\partial P_{bt}}]Q_{bt} = 0 \tag{12}$$

$$\frac{\partial \Pi_{bt}}{\partial \gamma_{bt}} = [(1+P_{bt})(1-F_{bt})+R_{bt}F_{bt}-C_{bt}]\frac{\partial Q_{bt}}{\partial \hat{\gamma}_{bt}} - \frac{\partial C_{bt}}{\partial \hat{\gamma}_{bt}}Q_{bt} = 0. \tag{13}$$

These two equations allow us to estimate, respectively, $\widehat{C}_{bt}$ and $\partial \widehat{C}_{bt}/\partial \hat{\gamma}_{bt}$ as:

$$\widehat{C}_{bt} = (1+P_{bt})(1-F_{bt})+R_{bt}F_{bt}+\left[1-F_{bt}+(R_{bt}-1-P_{bt})\frac{\partial F_{bt}}{\partial P_{bt}}\right]\frac{Q_{bt}}{\frac{\partial Q_{bt}}{\partial P_{bt}}} \tag{14}$$

and

$$\frac{\partial \widehat{C}_{bt}}{\partial \hat{\gamma}_{bt}} = \left[(1+P_{bt})(1-F_{bt})+R_{bt}F_{bt}-\widehat{C}_{bt}\right]\frac{\frac{\partial Q_{bt}}{\partial \gamma_{bt}}}{Q_{bt}}. \tag{15}$$

We can then recover c_{0bt} and c_{1bt} as follows:

$$\hat{c}_{1bt} = \frac{\frac{\partial \widehat{C}_{bt}}{\partial \hat{\gamma}_{bt}}}{\widehat{C}_{bt}}, \tag{16}$$

$$\hat{c}_{0bt} = \log(\widehat{C}_{bt}) - \hat{c}_{1bt}\hat{\gamma}_{bt}. \tag{17}$$

Finally, when consolidated, M&A banks in group j maximize the sum of savings banks' profits, $\sum_{b\in j}\Pi_{bt}$, with respect to P_{bt} and γ_{bt}. The ensuing optimality conditions allow us to recover c_{0bt} and c_{1bt} with consolidation. Based on the evidence of uncoordinated lending practices given in Section 3.2, we assume that SIP banks do not maximize the sum of profits when making their decisions and focus on the effects of consolidation via M&A.

4.2 Estimation

Our estimation proceeds in two steps. First, we use data on applications to estimate the acceptance model in Equation (6). We use a linear probability model to approximate for Equation (6), which allows us to include a large set of fixed effects, especially those controlling borrowers' demand for applications. To limit the impact of abrupt changes in the value of the estimated fixed effects, we define γ_{bt} as the maximum value of the acceptance rate over a window of 3 months. To capture credit screening at the bank level, we restrict our analysis to firms actively demanding new credit from banks with no previous relationship.

Specifically, we estimate credit supply shocks following Amiti and Weinstein (2018), but based on loan applications instead of credit growth

$$\phi_{ibt} = \gamma_{bt} + \underbrace{\rho Y_{it} + \zeta_{kst} + \eta_{mst} + \epsilon_{ibt}}_{\lambda_{ibt}}, \tag{18}$$

where Y_{it} are borrower financial variables (*Total liabilities/TA*, *Liquidity/TA*, *ROA*, and *log(TA)*); ζ_{kst} denote the interacted fixed effects for sector k (NACE industry), size bucket s (asset decile), and time t (month); and η_{kst} are the fixed effects for location m (province), size bucket s (asset decile), and time t (month). These fixed effects control for borrowers' demand for a loan, allowing us to interpret γ_{bt} as a supply-side decision on screening.[19] Finally, ϵ_{ibt} are borrower-bank-month specific normally distributed shocks.

We apply the following transformation to keep the logit functional form for Equation (6), giving us an acceptance probability between 0 and 1. Let $\widehat{\gamma}_{bt}$ be the estimated value of the rationing parameter from Equation (18). Then we compute the borrower-bank-month specific component, aggregated across borrowers, as:

$$\widehat{\lambda}_{bt} = \log\left(\frac{\phi_{bt}}{1-\phi_{bt}}\right) - \widehat{\gamma}_{bt}. \tag{19}$$

We will allow this aggregated borrower-bank-month component $\widehat{\lambda}_{bt}$ to remain constant across our counterfactuals. With this procedure, we obtain a ranking of acceptance decisions per bank-month combinations, which can change in the counterfactual analyses as banks adjust the rationing parameter γ . Although we use a linear model to estimate $\widehat{\gamma}_{bt}$ in place of the logit model, what matters for us is that, whatever the estimation procedure we use, it preserves the ranking of banks' acceptance decisions.[20]

In the second stage, we estimate the parameters of borrowers' indirect utilities from Equation (5) by simulated methods of moments. Our approach,

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19 We use borrower sector-size-time fixed effects and borrower location-size-time fixed effects because only 15% of the firms appear more than once in a given month (i.e., two or more banks request information on the creditworthiness of that firm in that month). Firm-time fixed effects used in Amiti and Weinstein (2018) would drastically reduce our sample size.

20 In a robustness exercise we estimate a variation of Equation (18) with both a linear probability model and a logit model. The logit model prevents the use of the same set of fixed effects used in Equation (18); consequently, we estimate a variation of Equation (18) that includes firm characteristics (instead of both firm characteristics and interacted fixed effects for borrowers' sector, size, and time as well as for borrowers' location, size, and time) to estimate demand and bank fixed effects (instead of bank-time fixed effects) to estimate time-invariant supply shocks. We find that the correlation between the coefficients for the bank fixed effects in both econometric specifications is 0.95. In addition, we investigated whether the ranking of the coefficients is similar for both methodologies. To this aim, we first order the two sets of fixed effects obtained under each methodology from the lowest to the highest value and define an ordinal variable with values spanning from 1 (corresponding to the bank with the lowest coefficient) and 255 (corresponding to the bank with the highest coefficient). Then, we take the absolute value of the difference between the ordinal values associated with each bank under each methodology. We find that the average absolute value of this difference is around one position, whereas the median is zero positions. Importantly, this difference is one position or less in more than 85% of the banks. This enables us to conclude that the rankings of bank fixed effects are very similar between the logit and linear probability models.

which combines demand estimation with endogenous choice sets, is close to Sovinsky Goeree (2008) and Abrams (2019). We simulate 200 borrowers $\imath$ for each bank-month combination, which all have the same acceptance probability $\phi_{\imath bt}$ within a bank-month combination, and draw a standard uniform random variable for each borrower-bank-month combination $u_{\imath bt}$. Each borrower will have bank b in their choice set if $\phi_{\imath bt} > u_{\imath bt}$, so their choice set will be defined as $\mathcal{S}_\imath$. Having the choice set of each simulated borrower allows us to calculate their choice probabilities as:

$$\mathcal{P}_{\imath bmt} = \frac{\exp(\delta^D_{bmt})}{1+\sum_{r\in\mathcal{S}_\imath}\exp(\delta^D_{rmt})}. \tag{20}$$

Finally, we calculate banks' market shares as $s_{bmt} = (\sum_{\imath=1}^{200}\mathcal{P}_{\imath bmt})/200$, where 200 is the number of borrowers we simulate.

In line with Berry et al. (1995), we then find the vector of δ^D_{bmt} such that the model market shares s_{bmt} equate the market shares observed in the data. This allows us to recover the structural error term $\xi^D_{bmt}(\delta^D,\theta^D) = \delta^D_{bmt} - \alpha^D P_{bt} - \beta^D X_{bt} - \tau^D_b - \omega^D_{mt}$, where $\theta^D = \{\alpha^D, \beta^D, \tau^D, \omega^D\}$, and construct the sample moments as $Z'\xi(\delta^D,\theta^D) = 0$, where Z are the instruments for interest rates. We use the lagged values of NPLs as an instrument for interest rates (P_{bt}). This choice aligns with Egan et al. (2017), who use lagged charge-offs in their deposit demand estimation exercise. Like charge-offs, lagged NPLs affect bank profitability for outstanding loans and, thus, loan rates on newly granted loans. Based on our tests, the instrument is relevant in the first stage, with the expected positive sign. It also satisfies the exclusion restriction, as past bank NPLs are likely to be unobserved by borrowing firms. This guarantees that they are uncorrelated with bank attributes ξ^D_{bmt} observed by borrowers but unobserved by the econometrician.[21]

We recover the parameters of borrowers' indirect utility from defaulting by estimating the following instrumental variables regression:

$$\log\frac{F_{bt}}{1-F_{bt}} = \alpha^F P_{bt} + \beta^F X_{bt} + \tau^F_b + \omega^F_t + \xi^F_{bt}. \tag{21}$$

As in the demand model, we instrument interest rates using the lagged values of NPLs.

4.3 Baseline results

For the estimation of the model, we use information relative to the new loans extended by all savings banks and the largest commercial and cooperative

[21] A possible concern of using NPLs as an instrument, despite being lagged, is that they could be correlated with borrowing firms performing poorly, influencing their demand for loans. To validate our results, we also perform an alternative estimation of the demand model using the amount of household deposits at the bank-month level as an instrument for loan interest rates, similar to Crawford et al. (2018). Being the primary source of bank funding, changes in deposit amounts represent a cost shock for banks' lending decisions. In this alternative specification, we find a demand elasticity of -2.16, which is very close to the ones recovered using NPLs as an instrument (-2.37).

Table 5
Summary statistics: Structural analysis

A. Descriptive statistics of the variables used for the estimation of demand

Variables	November 2007–November 2011					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
Market share (%)	2.49	0.39	4.62	<0.01	11.69	84,964
Interest rate (%)	4.59	4.52	1.28	2.45	6.61	3,119
Default risk (%)	7.95	6.96	4.09	4.03	16.34	3,119
TA (BE)	42.28	12.59	82.66	1.82	210.61	3,119

B. Firm and firm-bank variables used for the estimation of supply

Variables	November 2007–November 2011					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
TA (ME)	2.98	0.45	12.88	0.02	9.14	975,595
Total liabilities/TA (%)	68.06	75.66	29.06	8.64	100.00	975,595
Liquid assets/TA (%)	12.77	4.55	20.19	0.00	57.87	975,595
ROA (%)	4.42	5.55	21.24	−27.61	32.21	975,595
Accepted applications (%)	59.06	100.00	49.17	0.00	100.00	2,012,159

This table contains descriptive statistics (mean, median, standard deviation, 5th and 95th percentiles, and the number of observations) for bank characteristics (panel A) as well as for firm and firm-bank characteristics (panel B). In panel A, the variable *Market share* is aggregated at the bank-province-month level, and the variables *Interest rate* and *TA* are aggregated at the bank-month level. In panel B, we restrict the sample to firms that apply for new credit to banks with which they had no previous relationships. Firm characteristics (*TA*, *Total liabilities/TA*, *Liquid assets/TA*, and *ROA*) are aggregated at the firm-year level. In contrast, the share of accepted applications is aggregated at the firm-bank-month level. Panels A and B report the statistics between November 2007 and November 2011. For information on the construction of the variables, see the Internet Appendix.

banks for a total of 73 banks across 50 provinces, which together account for approximately 90% of the market. We aggregate the information on the remaining 274 banks in the outside option, which includes any credit entity with activity in Spain (many of which are credit cooperatives and other small credit institutions). The sample period goes between November 2007 and November 2011.

Table 5 provides descriptive statistics for the sample of banks, firms, and firm-bank relationships employed in estimating the structural model.[22] Panel A presents descriptive statistics for the variables used in the demand estimation, while panel B displays the variables employed in the supply estimation. In comparison to Table 1, the number of observations is different because here we use data from all banks and aggregate the information at the bank-province-month level (*Market share* in panel A) or the bank-month level (*Interest rate*, *Default risk*, and *TA* in panel A), and at the firm-year and bank-firm-month level (panel B). Moreover, in panel B, we restrict the sample to firms that apply for new credit to banks with which they had no previous relationships.

22 The notation we use for the market share at the 5th percentile in the table indicates that it is a positive number but very small.

Table 6
Demand and default estimation results

Variables	Demand	Default
Interest rate	–0.53***	1.02**
	(0.11)	(0.48)
log(TA)	1.16***	0.16
	(0.06)	(0.10)
Observations	84,964	3,119
Bank FE	Yes	Yes
Province-Month FE	Yes	Yes

An observation is a bank-month-province for the demand model and a bank-month for the default model. The demand model corresponds to Equation (7); the default model, to Equation (9). Robust standard errors are in parentheses for the demand model; standard errors are clustered at the month level for the default model. * $p<.1$; ** $p<.05$; *** $p<.01$. For information on the construction of the variables, see the Internet Appendix.

In contrast, in Table 1, we provide descriptive statistics for all banks' firm-bank relationships in which the lender is one of the groups involved in the consolidation process. We measure borrowers' default risk with the bank-month average probability of default within the next 12 months. We obtain the default risk of banks' portfolios by aggregating the probability of default of all their borrowers using the outstanding amount of credit of each firm in each bank as weights. The probability of default at the firm level is predicted based on the data from the CCR and CBSDO using the methodology outlined by Blanco et al. (2023) for Banco de España's internal credit assessment.[23]

In Table 6, we report the estimates of borrower demand and default determinants. We find that banks' market shares are increasing in size, as proxied by total assets, and that banks' expectations of borrower default risk are increasing in interest rates. The interest rate coefficient in the demand model implies a price elasticity of around -2.37, which is very close to other values obtained in the literature (Crawford et al. 2018). In Table 7, we provide the descriptives for the distribution of the cost parameters estimated using the entire sample period. Notably, we find that all values of $\hat{c}_{1bt}$ are positive, which means that a higher lending standards' parameter (corresponding to a larger number of accepted loan applications) raises the cost of lending because it rotates the average cost curve inward.

4.4 Counterfactual analyses

In our counterfactual analyses, we follow three steps. First, we simulate a scenario where we divest M&A groups and compare the model outcomes for banks with and without consolidation, assuming that consolidation generates no savings in either the operating or the screening cost component. This allows us to isolate the impact on credit conditions and bank performance of the market power effect achieved through credit policy coordination by M&A groups,

[23] According to the methodology, a firm is in default in a given year when it reports NPLs during at least 3 months of that year.

Table 7
Descriptives of cost parameters

Variables	Mean	Standard deviation	10th Percentile	Median	90th Percentile	N
$\hat{c}_{0bt}$	0.96	0.75	0.18	1.13	1.63	3,119
$\hat{c}_{1bt}$	0.29	0.79	0.03	0.11	0.42	3,119

This table contains descriptive statistics (mean, median, standard deviation, 5th and 95th percentiles, and the number of observations) for parameter c_{0bt}, which proxies banks' operating costs, and parameter c_{1bt}, which proxies banks' screening costs. See Equation (11). An observation is a bank-month.

Table 8
Divestiture versus M&A without and with cost efficiencies, lending standards

	Divested v. No cost efficiencies	M&A banks Cost efficiencies	M&A banks No lending standards
%Δ Credit quantity	6.41	3.22	−1.61
%Δ Interest rate	0.30	−7.53	2.05
Δ Acceptance rate	0.1	−9.64	−0.21
Δ Default risk	−0.11	−2.69	0.59
%Δ Profit	−6.61	−22.97	−1.13

The table reports the percentage changes (for *Credit quantity*, *Interest rate*, and *Profit*) and percentage point changes (for *Acceptance rate* and *Default risk*) in the equilibrium outcomes for the divested M&A savings banks relative to the same savings banks once they form the M&A groups (in the first two columns), which instead make coordinated decisions on interest rates and acceptance rates, assuming that M&A consolidation either does not produce any cost efficiency (in the first column) or that it does (in the second column). In the third column, we compare M&A banks optimizing only over interest rates to M&A banks optimizing over both interest rates and lending standards. To calculate the equilibrium outcome variables for the divested banks, since we do not observe their market shares in the postconsolidation period, we compute a weighted average across the banks that form each M&A group, using as weights the market share of each savings bank within the group as predicted by the model. We treat all the savings banks within each group j formed via a M&A as a single entity. An observation is at the bank group-month level. These results refer only to the months after the first consolidation (November 2009).

excluding any merger-related efficiency. The results are in the first column of Table 8. Second, to determine the total impact of consolidation on the variables of interest, we consider a counterfactual in which we compare a scenario with divested banks to one where M&A banks achieve savings in screening costs (second column of Table 8). Last, to quantify the importance of jointly modeling interest rates and lending standards, we present a counterfactual comparing M&A banks optimizing over both margins relative to M&A banks deciding only on interest rates.

4.4.1 The impact of market power absent cost savings. Table 8 reports the changes in the model outcomes due to the divestiture of M&A groups without merger-related cost efficiencies. We take the M&A groups resulting from consolidation operations and estimate their equilibrium outcomes, assuming they take coordinated choices. For the divested banks, since we do not observe their market shares in the postconsolidation period, we recover the market share of each bank in group j by solving for the optimal equilibrium outcomes they set when acting independently. This allows us to compute a weighted average of the outcome variables (*Credit quantity*, *Interest rate*, *Acceptance rate*, *Default risk*, *Profit*) across the banks in each group, using the market share predicted by the model as weights. We assume no change in c_{0bt} and c_{1bt} for both M&A and divested banks.

To remove any efficiency, all the relevant demand (ξ^D_{bmt}), acceptance (λ_{bt}), default (ξ^F_{bt}), and cost parameters (c_{0bt}, c_{1bt}) in the postconsolidation period, none of which are observed in the data, are predicted based on preconsolidation bank-specific trends. To perform this prediction, we regress each of the variables ($\xi^D_{bmt}, \xi^F_{bt}, \lambda_{bt}, c_{0bt}, c_{1bt}$) on bank and month fixed effects for the entire sample (all periods), excluding the information on M&A banks in the periods postconsolidation. We then predict the values of those variables for M&A banks postconsolidation using their bank fixed effects estimated in the preconsolidation period and the monthly fixed effects estimated across all banks in all periods. This means that we are assuming away the impact of postconsolidation cost efficiencies in both the scenario with M&A banks and the counterfactual scenario in which all M&A banks are divested.

The table reports the percentage (for *Credit quantity*, *Interest rate*, and *Profit*) and percentage point (for *Acceptance rate* and *Default risk*) differences in the equilibrium outcomes of the divested savings banks and the savings banks of M&A groups. We find that divested banks increase their credit quantity compared to when they are part of a M&A group. This is due to a slight increase in divested banks' acceptance rates, compensating for the slight rise in interest rates compared to M&A banks. Additionally, divested banks reduce the ex ante risk they take, thus lowering the default risk of their borrowers. Overall, profits decrease for divested banks compared to M&A banks. We attribute these results to the market power effect of M&A consolidation.

4.4.2 The role of screening costs' savings. We investigate the differences in the values of c_{1bt} with consolidation and efficiencies as opposed to those without consolidation. For this analysis, we predict the value of c_{1bt} without cost efficiencies as described above. We plot their distributions in Figure 2. We use these values in our counterfactual analysis to study the impact of efficiencies on the equilibrium outcomes obtained in the case of consolidation and efficiencies and in the case of bank divestiture. We observe a significant shift toward lower values of c_{1bt} for M&A banks compared to divested banks. The divergence between the two distributions documents the role of consolidation in enhancing banks' ability to lower screening costs. We also find that the two-sample Kolmogorov-Smirnov test for equality of distribution functions, comparing the two distributions of c_{1bt}, rejects the null hypothesis of equal distributions with a K-S test statistic of 0.4306 and a p-value of .000. We interpret this as evidence that M&A consolidation reduces bank screening costs to firms of any risk. This can be achieved through improvements in firm monitoring or the capability to process the information on borrowers' credit merits, as shown by Panetta et al. (2009).

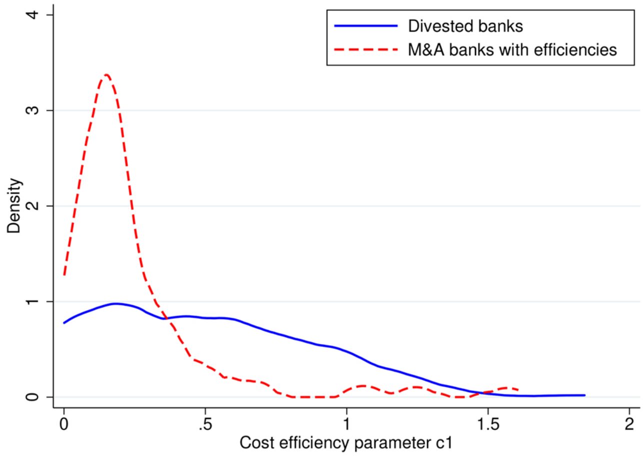

Figure 2
Cost parameter c_{1bt} with and without consolidation
The two-sample Kolmogorov-Smirnov test for equality of distribution functions, comparing the distribution of c_{1bt} with and without merger, rejects the null hypothesis of equal distributions with a K-S test statistic of 0.4306 and a p-value of .000.

4.4.3 The total impact of M&A consolidation on credit supply and performance. The findings in the first column of Table 8 show that M&A consolidation reduces the quantity of lending in the economy. Another force we unveil through the model is that M&A consolidation generates savings in screening costs (Figure 2), enabling M&A banks to become more efficient at lending to riskier firms. To determine how these effects balance, we consider a counterfactual analysis in which we compare the equilibrium outcomes for divested banks to the equilibrium outcomes for M&A banks when they benefit from savings in screening costs. The results are in the second column of Table 8.

The magnitude of the changes in equilibrium outcomes in columns 1 and 2 is remarkably different, underscoring the importance of the screening cost efficiencies documented in Figure 2. Compared to divested banks, the credit supply of merged banks falls because the increase in their acceptance rates does not compensate for the contractionary effect of their higher interest rates. We also find that the reduction in screening costs and the rise in interest rates imply that merged banks are induced to extend credit to ex ante riskier firms. Given these results, the question is whether the arguably negative impact of a higher ex ante risk taken by consolidated banks can be compensated by the significant improvement in their efficiency in screening borrowers. We then turn to the analysis of banks' profitability. When accounting for screening cost efficiencies, we show that profits are significantly bigger for merged banks than for divested banks. That is, the combined impact of market power and screening cost reductions produces an increase in M&A banks' profits compared to divested banks.

4.4.4 The impact of modeling lending standards on M&A banks' outcomes. To document the importance of modeling lending standards, we run a simulation comparing the scenario where M&A banks optimize only over interest rates to one where they optimize over both interest rates and acceptances. The results are in the third column of Table 8. This counterfactual assumes that M&A consolidation produces the same screening cost efficiency in both scenarios. Thus, this exercise allows us to isolate the impact of enabling banks to adjust their lending standards by changing their acceptance rates postmerger.

The results suggest that if merged banks decide on both margins, their interest rates are lower, leading to lower defaults. Moreover, acceptance rates are higher. Lower interest rates and higher acceptance rates produce larger lending volumes for consolidated banks that can optimize their lending standards. The possibility of using an additional margin to screen borrowers allows these banks to extend credit to safer borrowers without raising interest rates, which explains the lower default risk.

Overall, the profits of merged banks that adjust both margins are higher than those of merged banks that can set only their interest rates. This result underscores the importance of employing a model that more accurately reflects banks' decision-making processes to predict the effects of consolidation in the banking sector.

4.5 Validation of the model results

The purpose of our equilibrium model is the comparison between merged banks and their counterfactual divestiture. To validate the results of this exercise, we rely on the reduced-form evidence of Section 3, where we compare merged banks to business groups, with the latter resembling the divestiture scenario of the model. The results of the second column of Table 8 are consistent with the first three columns of Table 4. In both cases we find that merged banks increase interest rates and acceptance rates while experiencing a decline in the amount of credit originated.

The results on credit performance cannot instead be directly compared across the two tables because the reduced form analysis uses NPLs to examine ex post risk. The structural model uses banks' expectations of borrowers' default to quantify ex ante risk. Table 8 shows that M&A banks, despite taking more risk than when divested, are more profitable than divested banks. We interpret this result as a consequence of improved screening practices. However, the results of Table 4 do not explore the implications of more efficient screening due to bank consolidation for ex ante risk. To bridge the reduced-form and the structural analyses, we empirically analyze the evolution of firms' NPL based on the ex ante risk of banks' loan portfolios.

We estimate Equation (4) in subsamples of firms categorized by their ex ante risk levels and report the results in Table 9. Columns 1 and 2 display the results for firms with a probability of default at the time of loan issuance that is either

Table 9
Credit risk borne by M&A banks

Variables	(1) PD≤Median	(2) PD>Median	(3) PD>3%	(4) PD>5%
M&A	–0.005*	–0.016***	–0.031**	–0.057***
	[0.003]	[0.006]	[0.013]	[0.022]
Observations	45,115	44,925	14,059	4,656
Industry-Location-Size-Time FE	YES	YES	YES	YES
Bank Controls	YES	YES	YES	YES
Firm Controls	YES	YES	YES	YES

This table reports the results obtained from four instrumental variable regression analyses in which the instrument is a binary variable that indicates whether the credit overlap in the bank group region of influence is high. The dependent variable in columns 1-4 is the average proportion of NPLs at the end of every semester in the period spanning between the announcement of group j-formation (M&A or SIP) and November 2011. This regression analysis is conducted for various firms' subsamples categorized by risk levels. Columns 1 and 2 display results for firms with a probability of default at the time of loan issuance either below or above the median credit risk distribution for the sample. In columns 3 and 4, we narrow the sample to two groups of even riskier firms. Column 3 includes firms with a probability of default above 3%, corresponding to a credit quality step (CQS) category defined by the ECB of 7. In contrast, column 4 focuses on firms in the highest (riskiest) CQS, which is 8. The explanatory variable of interest is the dummy *M&A*, which is estimated only for the postconsolidation period since we consider the bank-firm pairs with no credit relationship as of the beginning of the consolidation period (i.e., November 2009). The set of control variables in all columns includes bank characteristics such as *log(TA)*, *Capital ratio*, *ROA*, and *FROB/TA*. Additionally, we use industry-location-size-time fixed effects and the following firm characteristics as control variables: *Total liabilities/TA*, *Liquidity/TA*, *ROA*, and *log(TA)*. Robust standard errors (in brackets) are clustered at industry-province-size and bank group-province level. * $p<.1$; ** $p<.05$; *** $p<.01$. For additional information, see the Internet Appendix.

below or above the median credit risk distribution for the sample. We use SIPs to quantify the impact of consolidation on the variable of interest. Compared to SIP banks, M&A banks report fewer NPLs across firm risk categories, with a more significant decrease for riskier firms. That is, the higher ex ante risk taken by M&A banks after consolidation, due to fewer rejected applications, does not translate into higher ex post risk. This explains the increase in the profitability of consolidated banks in Table 8.

In columns 3 and 4, we narrow the sample to two groups of even riskier firms. Column 3 includes firms with a probability of default above 3%, corresponding to a credit quality step (CQS) category of 7 as defined by the European Central Bank (ECB). Column 4 focuses on firms in the highest (riskiest) CQS (equal to 8). The results indicate that, compared to SIP banks, M&A banks postconsolidation had more effective screening procedures that allowed them to take on more risk without compromising the credit quality of their portfolios.

5. Conclusions

Between 2009 and 2011, the Spanish banking system underwent a restructuring process to consolidate its savings banks' sector. We exploit the institutional design of the consolidation program to study how banks' consolidation affects credit supply and performance. We propose a quasi-experimental analysis based on a new identification strategy. It shows that bank mergers restrict credit supply, set higher interest rates, reject fewer applicants, and report fewer

NPLs. To unveil the equilibrium forces behind these results, we propose a new structural model of demand and supply of credit, in which differentiated banks set interest rates and lending standards. We find that, despite a relaxation in lending standards, the increase in interest rates determined a reduction in merged banks' credit quantity. Moreover, merged banks' profitability increases despite extending credit to borrowers with higher default risk. We attribute these results to the combined effect of merger market power and the significant drop in merged banks' screening costs.

The validity of our analysis extends beyond the Spanish case. Governments and regulators give the green light to bank consolidation programs in times of crisis as a response to the problems of ailing banks (Corbae and Levine, 2018). After the recent financial crisis, the banking sector of many countries was significantly affected by restructuring measures similar to those decided by the Spanish government (European Commission, 2017). In the United States, the FDIC auction process was used after the crisis to resolve insolvent banks, resulting in a consolidation process induced by the regulator (Allen et al. 2024). In Japan, after the NPL crisis of the late 1990s, the government injected public capital into the banking sector and advised banks to merge (Hoshi and Kashyap, 2004). Policy makers claimed that consolidation could reduce the inclination of "overbanked" systems leading to excessive NPL stockpiling in times of crisis.[24] The debate during the COVID-19 pandemic made no difference.[25] On the one hand, we find that consolidation leads to a reduction in NPLs. On the other hand, our model shows that the improvement in banks' performance comes from merged banks becoming more efficient in screening borrowers while contemporaneously relaxing their lending standards.

Code availability: The replication code is available in the Harvard Dataverse at https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/JFEZXC

References

Abrams, E. 2019. Assessing bank deposit market power given limited consumer consideration. SSRN 3431374.

Adams, W., L. Einav, and J. Levin. 2009. Liquidity constraints and imperfect information in subprime lending. *American Economic Review* 99:49–84.

Aguirregabiria, V., R. Clark, and H. Wang. 2024. The geographic flow of bank funding and access to credit: Branch networks, local synergies and competition. Working Paper, University of Toronto.

Allen, J., R. Clark, and J.-F. Houde. 2014. The effect of mergers in search markets: Evidence from the Canadian mortgage industry. *American Economic Review* 104:3365–3396.

[24] See, for example, "Bank Competition and Bank Supervision" by Ignazio Angeloni (Member of the ECB Supervisory Board at the time), 4 July 2016, available at https://www.bankingsupervision.europa.eu/press/speeches/date/2016/html/se160704.en.html.

[25] See "Consolidation in the European banking sector: challenges and opportunities" by Edouard Fernandez-Bollo (Member of the ECB Supervisory Board), 11 June 2021, available at https://www.bankingsupervision.europa.eu/press/speeches/date/2021/html/ssm.sp210611~87256e1f4b.en.html.

Allen, J., R. Clark, B. Hickman, and E. Richert. 2024. Resolving failed banks: Uncertainty, multiple bidding & auction design. *Review of Economic Studies* 91:1201–1242.

Amiti, M., and D. E. Weinstein. 2018. How much do idiosyncratic bank shocks affect investment? Evidence from matched bank–firm loan data. *Journal of Political Economy* 126:525–587.

Banco de España. 2015. Nota informativa sobre ayudas públicas en el proceso de reestructuración del sistema bancario español (2009–2015).

——— 2017. Informe sobre la crisis financiera y bancaria en España, 2008–2014.

Benetton, M. 2021. Leverage regulation and market structure: A structural model of the UK mortgage market. *Journal of Finance* 76:2997–3053.

Berger, A. N., A. Saunders, J. Scalise, and G. Udell. 1998. The effects of bank mergers and acquisitions on small business lending. *Journal of Financial Economics* 50:187–229.

Berry, S., and A. Pakes. 1993. Some applications and limitations of recent advances in empirical industrial organization: Merger analysis. *American Economic Review* 83:247–252.

Blanco, R., E. Fernandez, M. Garcia-Posada, and S. Mayordomo. 2023. An estimation of default probabilities of Spanish non-financial corporations and its application to evaluate public policies. *Banco de España Occasional Papers* No. 2319.

Bonaccorsi di Patti, E., and G. Gobbi. 2007. Winners or losers? The effects of banking consolidation on corporate borrowers. *Journal of Finance* 62:669–695.

Clark, R., J.-F. Houde, and J. Kastl. 2021. The industrial organization of financial markets. In *Handbook of Industrial Organization*, eds. K. Ho, A. Hortasçu, and A. Lizzeri, 427–520. Amsterdam: Elsevier.

Corbae, D., and R. Levine. 2018. Competition, stability, and efficiency in financial markets. Jackson Hole Symposium: *Changing Market Structure and Implications for Monetary Policy.*

Crawford, G. S., N. Pavanini, and F. Schivardi. 2018. Asymmetric information and imperfect competition in lending markets. *American Economic Review* 108:1659–1701.

Crawford, G. S., O. Shcherbakov, and M. Shum. 2019. Quality overprovision in cable television markets. *American Economic Review* 109:956–995.

Cuesta, J. I., and A. Sepulveda. 2021. Price regulation in credit markets: A trade-off between consumer protection and credit access. SSRN 3282910.

Degryse, H., O. De Jonghe, S. Jakovljevic, K. Mulier, and G. Schepens. 2019. Identifying credit supply shocks with bank–firm data: Methods and applications. *Journal of Financial Intermediation* 40.

Degryse, H., N. Masschelein, and J. Mitchell. 2011. Staying, dropping or switching: The impact of bank mergers on SMEs. *Review of Financial Studies* 24:1102–1140.

Egan, M., A. Hortaçsu, and G. Matvos. 2017. Deposit competition and financial fragility: Evidence from the U.S. banking sector. *American Economic Review* 107:169–216.

Einav, L., M. Jenkins, and J. Levin. 2012. Contract pricing in consumer credit markets. *Econometrica* 80:1387–1432.

Erel, I. 2011. The effect of bank mergers on loan prices: Evidence from the United States. *Review of Financial Studies* 24:1068–1101.

European Commission. 2017. Coping with the international financial crisis at the national level in a European context. European Commission Staff Working Document, SWD(2017)373.

Favara, G., and M. Giannetti. 2017. Forced asset sales and the concentration of outstanding debt: Evidence from the mortgage market. *Journal of Finance* 72:1081–1118.

Focarelli, D., and F. Panetta. 2003. Are mergers beneficial to consumers? Evidence from the market for bank deposits. *American Economic Review* 93:1152–1172.

Fuentelsaz, L., and J. Gomez. 2001. Strategic and queue effects on entry in Spanish banking. *Journal of Economics & Management Strategy* 10:529–563.

Giannetti, M., and F. Saidi. 2019. Shock propagation and banking structure. *Review of Financial Studies* 32:2499–2540.

Granja, J., G. Matvos, and A. Seru. 2017. Selling failed banks. *Journal of Finance* 72:1723–1784.

Guiso, L., P. Sapienza, and L. Zingales. 2004. Does local financial development matter? *Quarterly Journal of Economics* 119:929–969.

Hoshi, T., and A. K. Kashyap. 2004. *Solutions to Japan's banking problems: What might work and what definitely will fail*. Stigler Center Working Paper.

Houston, J. F., C. M. James, and M. D. Ryngaert. 2001. Where do merger gains come from? Bank mergers from the perspective of insiders and outsiders. *Journal of Financial Economics* 60:285–331.

International Monetary Fund. 2012. Spain: The reform of Spanish savings banks. IMF Country Report No. 12/141.

Jiménez, G., S. Ongena, J.-L. Peydró, and J. Saurina. 2014. Hazardous times for monetary policy: What do 23 million loans say about the impact of monetary policy on credit risk-taking? *Econometrica* 82:463–505.

Lester, B., A. Shourideh, V. Venkateswaran, and A. Zetlin-Jones. 2019. Screening and adverse selection in frictional markets. *Journal of Political Economy* 127:338–377.

Liberti, J. M., and M. A. Petersen. 2019. Hard and soft information. *Review of Corporate Finance Studies* 8:1–41.

Lim, P. 2020. Interest rates and credit rationing in U.S. mortgage lending. Unpublished manuscript.

Méndez Álvarez-Cedrón, J. M. 2011. *Cajas de ahorros: Nueva normativa*. Fundación Cajas Ahorros Confederada: Madrid.

Panetta, F., F. Schivardi, and M. Shum. 2009. Do mergers improve information? Evidence from the loan market. *Journal of Money, Credit and Banking* 41:673–709.

Peek, J., and E. S. Rosengren. 1998. Bank consolidation and small business lending: It's not just bank size that matters. *Journal of Banking and Finance* 22:799–817.

Petersen, M. A., and R. G. Rajan. 1995. The effect of credit market competition on lending relationships. *Quarterly Journal of Economics* 110:407–443.

Robles-Garcia, C. 2022. Competition and incentives in mortgage markets: The role of brokers. Working Paper No. 3796, Stanford Business.

Sapienza, P. 2002. The effects of banking mergers on loan contracts. *Journal of Finance* 57:329–367.

Scharfstein, D. S., and A. Sunderam. 2016. Market power in mortgage lending and the transmission of monetary policy. Harvard University Working Paper.

Sovinsky Goeree, M. 2008. Limited information and advertising in the U.S. personal computer industry. *Econometrica* 76:1017–1074.

Starc, A. 2014. Insurer pricing and consumer welfare: Evidence from Medigap. *Rand Journal of Economics* 45:198–220.

Stein, J. 2002. Information production and capital allocation: Decentralized versus hierarchical firms. *Journal of Finance* 57:1891–1921.

Stiglitz, J. E., and A. Weiss. 1981. Credit rationing in markets with imperfect information. *American Economic Review* 71:393–410.

Wang, Y., T. M. Whited, Y. Wu, and K. Xiao. 2022. Bank market power and monetary policy transmission: Evidence from a structural estimation. *Journal of Finance* 77:2093–2141.